THE IDEAL CITY OF THE FUTURE

ISSUE 01 · 2026

# AURORA HAVEN

A POLAR CITY THAT THRIVES



## Its own sun. Its own food.
## A mountain you can walk inside.

INSIDE −45 °C and no sun: the polar challenge p. 6 · Explore our 3D city p. 21 · Invest in Aurora Haven p. 32

POLAR CITY · CLASS 11A4 · LÊ QUÝ ĐÔN HIGH SCHOOL

A CONCEPT MAGAZINE



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# *“Our goal was not only to survive the cold but to build a city where people really want to live.”*



*The Aurora Lagoon, an open-air mineral pool at the foot of Mount Aurora, under the northern lights.*

**WELCOME TO THIS SPECIAL FEATURE** on a polar city named Aurora Haven – a bold architectural marvel where human ingenuity meets the call of the wild. In an era shaped by global climate shifts, this project is more than just an urban concept; it represents a strategic vision for survival and sustainable development in the earth’s most extreme polar environments.

From its sleek structural design to intelligent renewable energy integration, Aurora Haven redefines how humanity can adapt and thrive under pressure. We invite you to turn the page and step into a world where extreme natural challenges are transformed into a vibrant, resilient, and forward-thinking community.

***The Polar City team***

Class 11A4 · Lê Quý Đôn High School



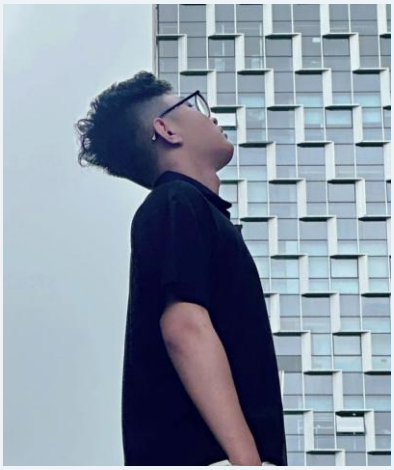
ON THE COVER

The whole of Aurora Haven from the air on a polar night: downtown, the stadium, the greenhouse dome and the plants, under the northern lights.

THE TEAM BEHIND AURORA HAVEN

# Meet the Founders

*Who we are, and what each of us did.*



### Nguyễn Đông Bảo

**Magazine design · 3D city design**



### Nguyễn Bá Hiếu Thiện

**Team leader · Research · 3D city design**



### Phan Minh Khôi

**Research · Fact-checking**



### Dương Hàng Nguyên

**Video filming & editing · 3D city design**



### Phạm Phước Khoa

**Video editing · Magazine design**



### Nguyễn Minh Triết

**Magazine design · Report**

SCAN TO WATCH

**Our real-world interview**

Video on YouTube
youtu.be/QwcP_2Q3Esc

INTERVIEW CONCLUSION

*Looking back on the interview session, our team not only gathered valuable architectural insights but also learned a great deal about creative thinking and teamwork. Preparing the questions, listening closely, and engaging in direct discussions helped us deeply understand how green technologies can be practically applied to extreme environmental challenges. Overall, this was a truly rewarding experience that broadened our perspectives and allowed us to bring the most inspiring stories to our magazine readers.*



CITY OVERVIEW

# Welcome to Aurora Haven

*A compact, zero-carbon city, built to stay warm and happy through the polar night.*

**AURORA HAVEN IS** a city for about 45,000 people on the Arctic tundra, under the snow peak of Mount Aurora.

Everything is a short walk away. One heating network warms every building, and almost all water and waste is used again.

*"Three big ideas work as one loop."*

**PROJECT DATA**

| | |
|---|---|
| LOCATION | **Arctic tundra, about 70° North** |
| POPULATION | **About 45,000** |
| CLIMATE | **Winters to −45 °C; two months of polar night** |
| STATUS | **A concept for the future** |

*Aurora Haven at dusk from the south-east, with the wind farm on the northern edge.*

## THREE BIG IDEAS



1 **The City's Own Sun** PAGE 12

A fusion reactor powers and heats every home.



2 **Bread and Water from Snow and Air** PAGE 15

A plant turns snow and air into water and food.



3 **A Snow Mountain You Can Walk Inside** PAGE 17

Warm pools and saunas inside a snow peak.



THE PLACE

# The Polar Challenge

*One of the hardest places on Earth to build a city.*

Aurora Haven stands on the Arctic tundra at about 70° North. Winter falls to −45 °C, and the sun stays down for about two months, as in Tromsø, Norway [1].

The ground is permafrost: a warm building can thaw it and crack. The Arctic is warming nearly four times faster than the world [2], and much of its infrastructure is at risk by 2050 [3].

With no farmland, winter supplies come by air. So we asked one question about every building:

***"How does it help people stay warm, fed, safe and happy here?"***

**−45 °C**

in winter

**2 months**

of polar night

**4×**

faster warming than the world

*A blizzard sweeps past planted spruce windbreaks.*



*Downtown in the low sun.*

THE VISION

# Our Vision & Mission

***Our mission: a zero-carbon city where 45,000 people stay warm, fed, safe and happy through the polar night, without harming the tundra.***

## WHO IS AFFECTED

### Arctic communities

About 4 million people live in the Arctic [4], many on thawing ground [3].

### Supply chains

In winter, food and medicine come only by air. One long storm can cut a town off.

### The tundra

Foxes, birds and reindeer need frozen land that a careless city could melt.

## THE CRISIS, AND OUR ANSWER

| | | |
|---|---|---|
| **Cold down to −45 °C** | → | One shared heating loop and super-insulated homes |
| **Two months of darkness** | → | Bright public spaces and a warm spa mountain |
| **No farmland** | → | Water and food made from snow and air |
| **Thawing ground** | → | Insulated piles and thermosyphons keep it frozen |
| **Cut off by storms** | → | Our own power, shelters and blizzard warnings |



DOWNTOWN

# Nine Blocks, One Short Walk

*Shops, schools, healthcare and homes, all a short walk from the Founding Mayor Monument.*

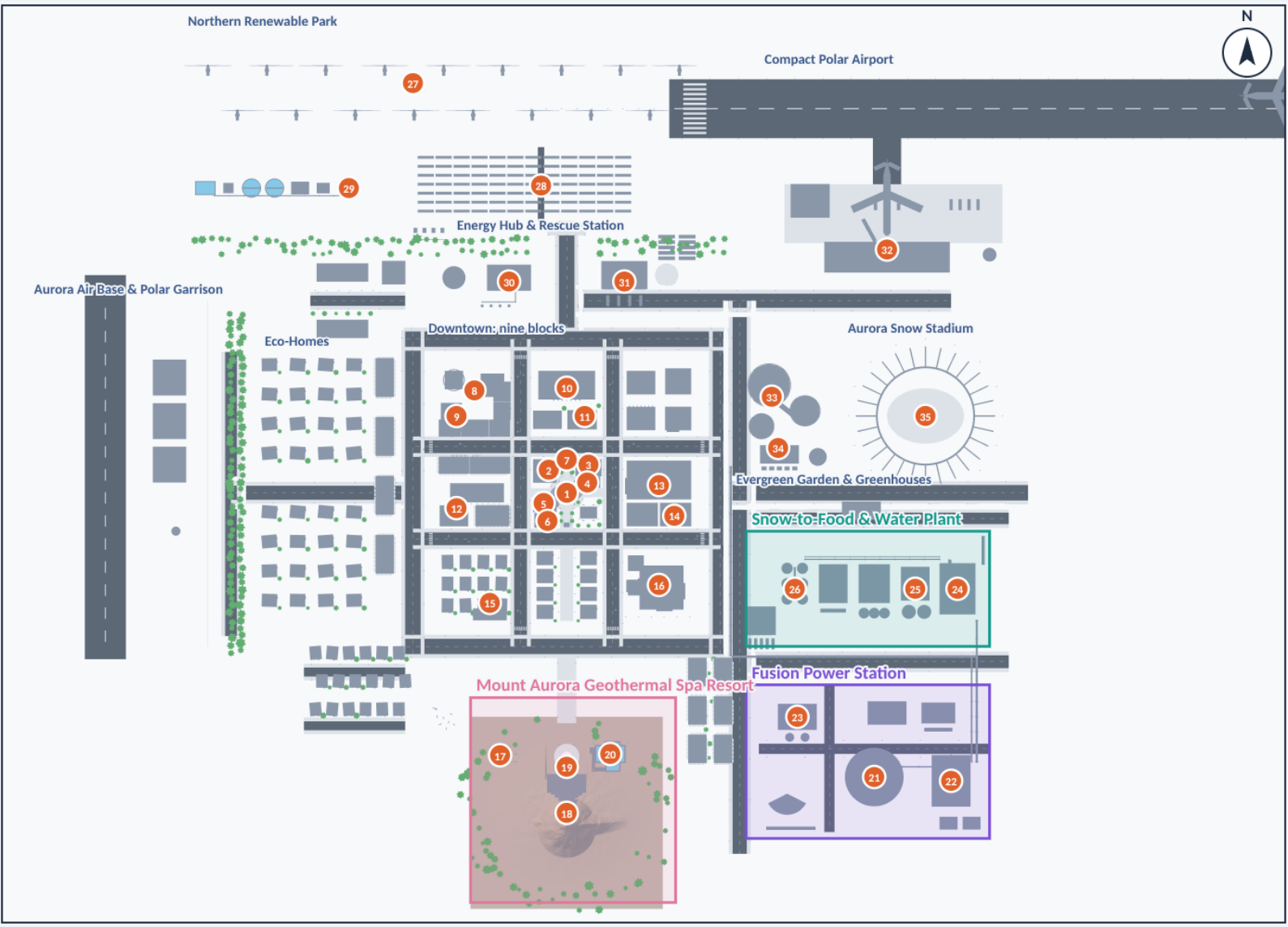
Liberty Mall, the pedestrian heart of downtown, hosts markets and festivals.



*Liberty Mall at dusk, looking north to the Founding Mayor Monument.*

THE MAP

# The Whole City Plan




*An illustrative plan from our 3D model, not to scale. North is up; the numbers are the places on page 34.*

### DOWNTOWN

Nine blocks around the Founding Mayor Monument.

### NORTH

Wind and solar farms, the rescue station and the airport.

### WEST

Eco-homes behind a planted tree belt.

### EAST

The fusion station, the Snow-to-Food plant and the greenhouses.

### SOUTH

Mount Aurora, with its spa inside.

DOWNTOWN

# The Nine Blocks



*Downtown from the south-east; the cards follow the city plan.*

**Polaris Quarter**

AI control tower · police

**Knowledge Quarter**

University · library

**Financial District**

Stock exchange · courthouse

**West Market Street**

Fire station · night market

**Civic Core & Liberty Mall**

Monument · hospital · school · metro

**East Gallery Avenue**

Glacier Mall · music hall

**Snowpine Residences**

Family homes · senior care

**Mount Aurora Approach**

Guesthouse · spa shuttle

**Aurora 81 District**

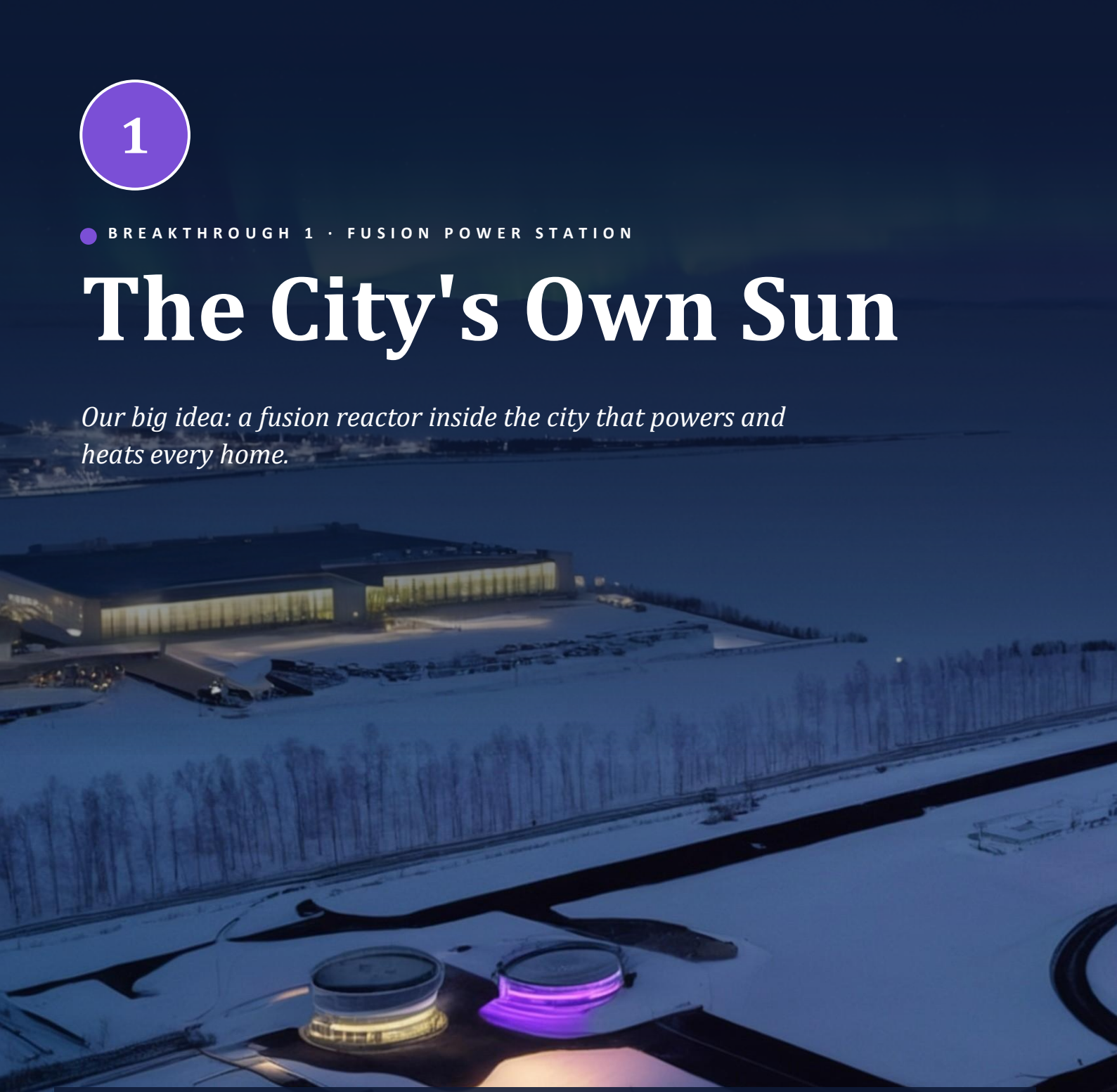
Landmark tower · twin towers

1

BREAKTHROUGH 1 · FUSION POWER STATION

# The City's Own Sun

*Our big idea: a fusion reactor inside the city that powers and heats every home.*



## IN BRIEF

### THE PROBLEM

Two months of polar night at −45 °C. The city needs heat and power that never stop.

### THE IDEA

A compact fusion reactor makes the city's electricity and heat, with fuel from snowmelt.

### WHAT'S NEW

A floating fission plant already heats the Arctic town of Pevek, Russia [5]. Ours is fusion, fuelled by snowmelt, with no long-lived waste [6].

### THE IMPACT

No fuel to ship in and no $CO_2$. If anything fails, the reaction simply stops.

WHERE THE WORLD IS TODAY

No fusion plant sells power yet. ITER plans to burn fusion fuel in 2039 [7], and breeding tritium is still to be tested [8].

REALITY CHECK

## ≈ 0.2 kg a day

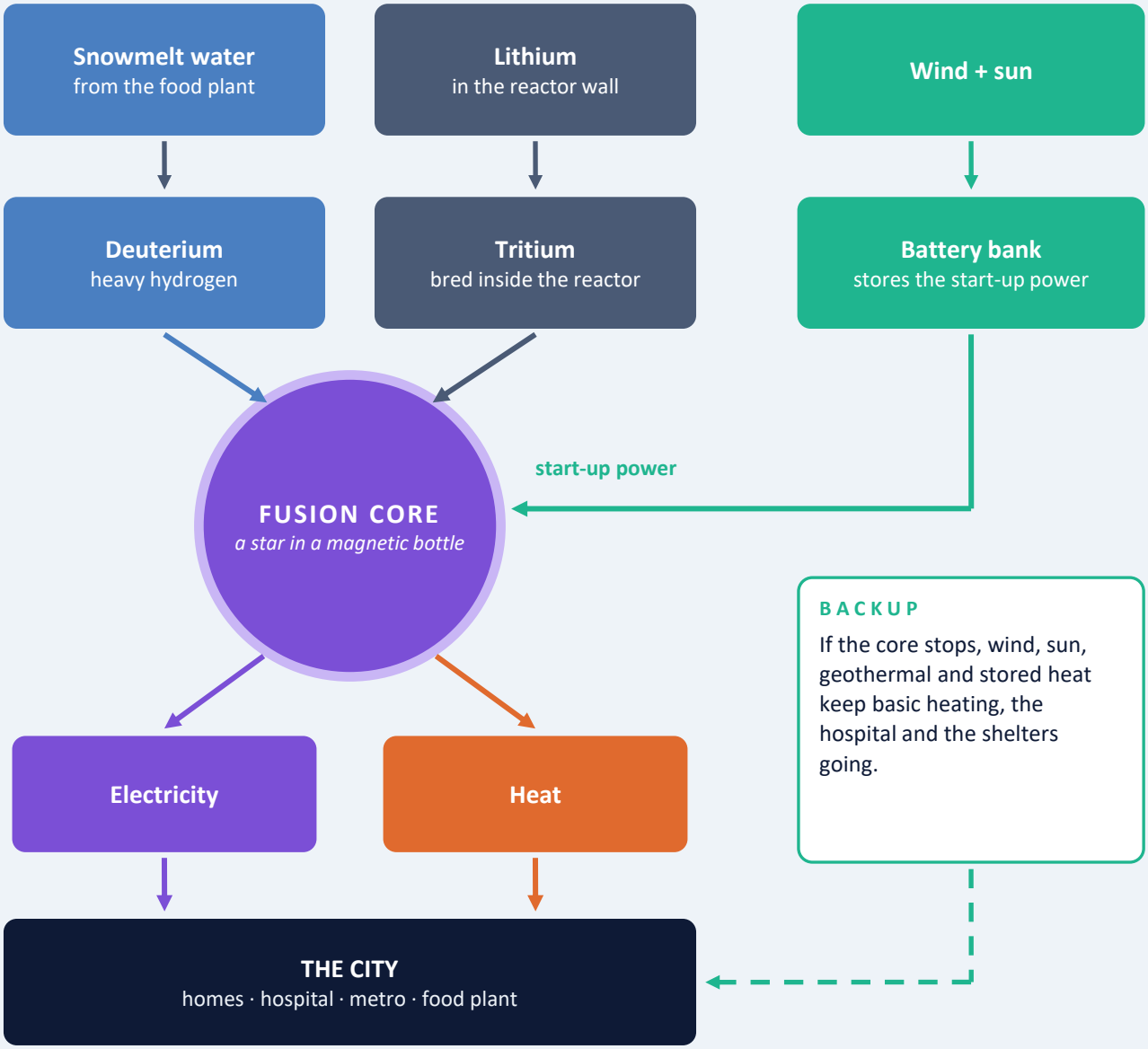
Our estimate: all the fuel for a full day of city power and heat. Until fusion is proven, wind, sun and geothermal run the city (page 31).



*The Fusion Core under the northern lights.*

# How Our Sun Works

FROM SNOW TO POWER AND HEAT




## Fuel from snow

Deuterium comes from snowmelt; tritium is made from lithium in the reactor wall [6].

## Safe by design

If anything goes wrong, the plasma cools in seconds. Tritium stays in sealed loops.

## Nothing burns

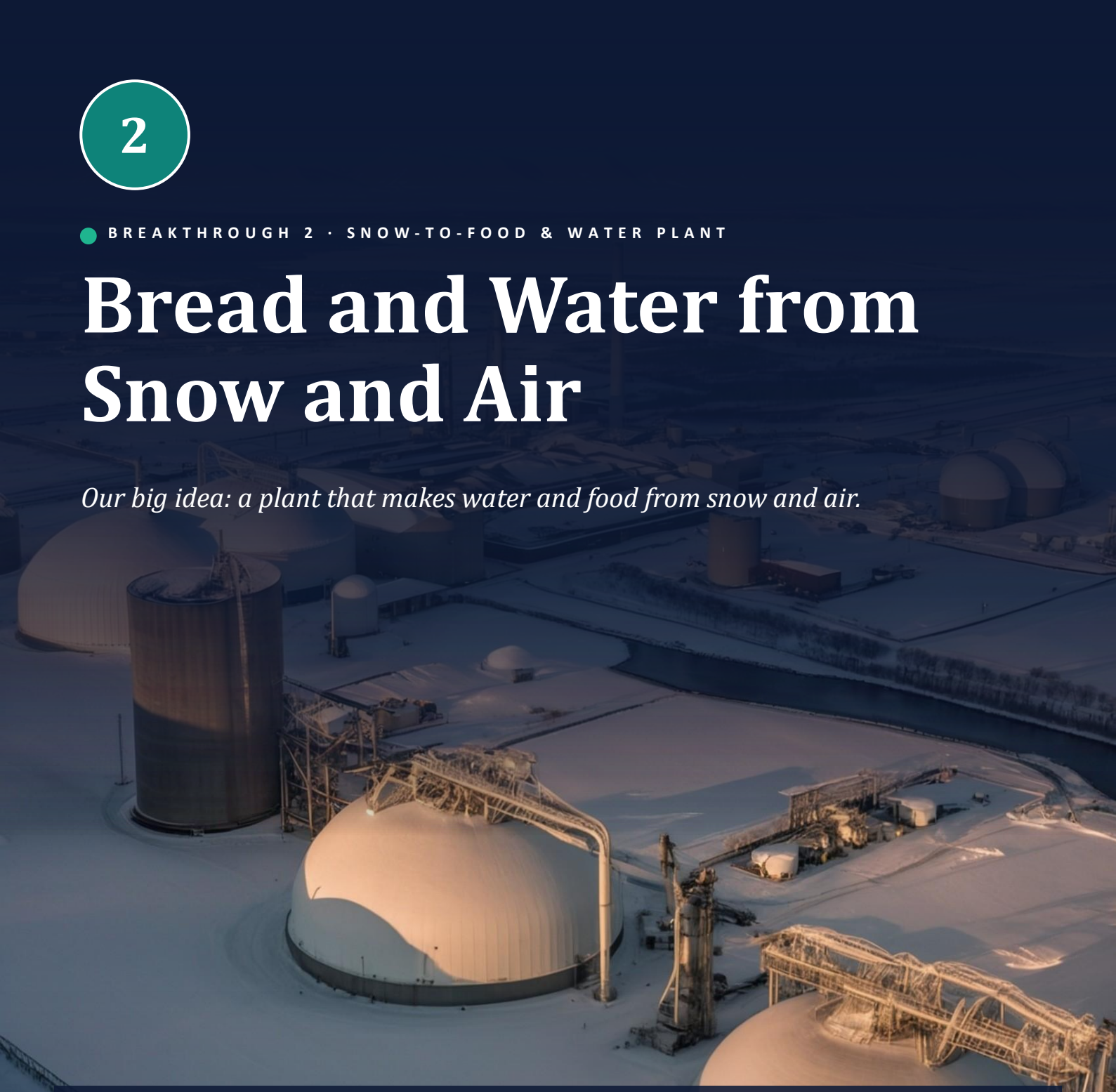
No $CO_2$, and the reactor's spare heat melts the snow we drink.

2

BREAKTHROUGH 2 · SNOW-TO-FOOD & WATER PLANT

# Bread and Water from Snow and Air

*Our big idea: a plant that makes water and food from snow and air.*



IN BRIEF

*Above: the Bioreactor Silos in the low sun.*

### THE PROBLEM

Almost nothing grows at −45 °C, and winter food comes only by air.

### THE IDEA

Spare fusion heat melts snow for drinking water; microbes turn hydrogen and $CO_2$ into protein flour.

### WHAT'S NEW

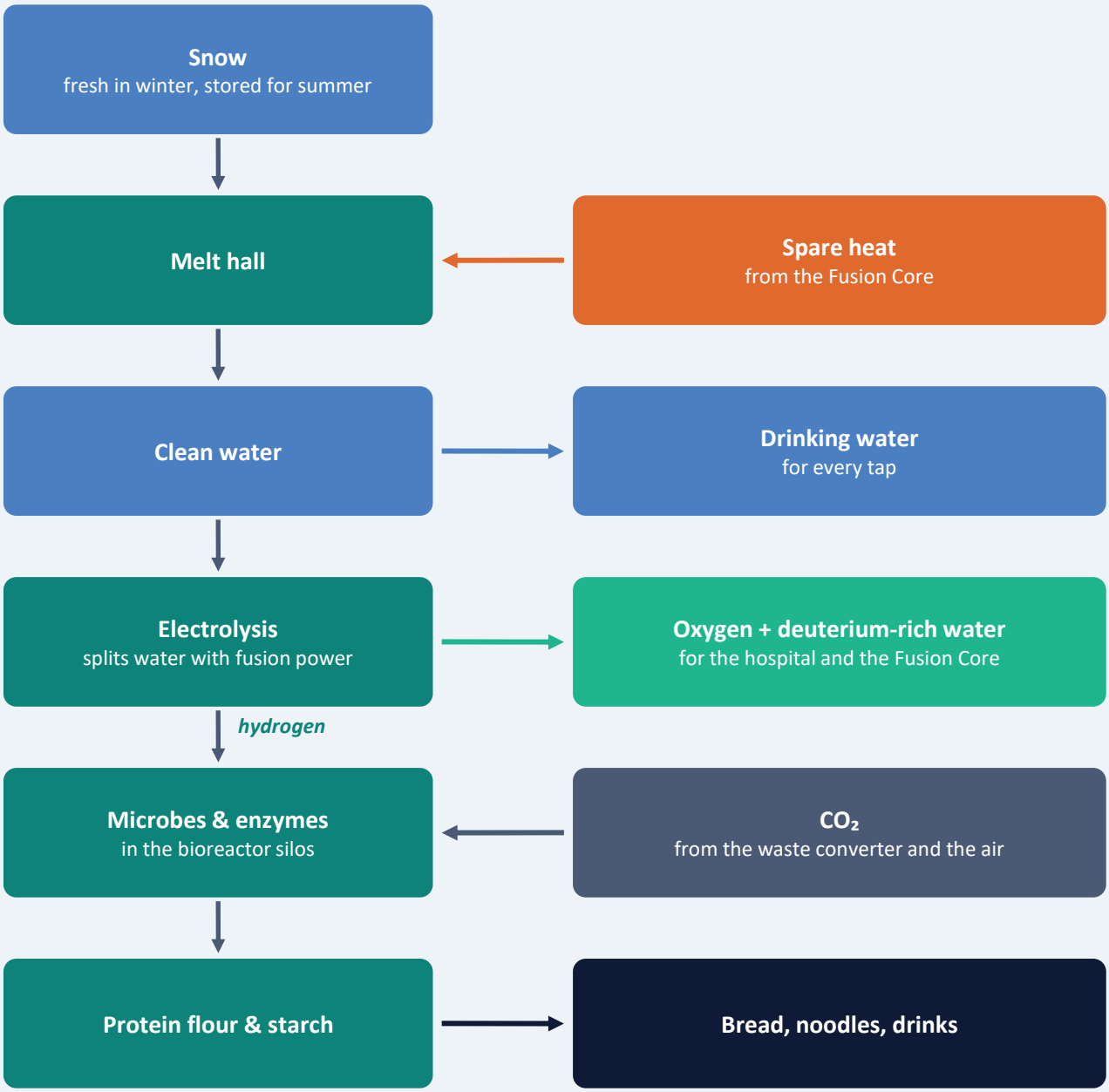
Solar Foods makes protein from air [9], and Denmark's Kalundborg factories share heat and water [10]. We join both in one closed loop inside a city.

### THE IMPACT

Bread, noodles and drinks made here all year, with fewer supply flights.

# From Snow to Supper

FROM SNOW AND AIR TO WATER AND FOOD




## WHERE THE WORLD IS TODAY

Solar Foods in Finland already makes Solein protein from air and electricity [9]. Starch from $CO_2$ works, but only in a lab so far [11].

## REALITY CHECK

Snow does not fall all year, so winter snow and ice are stored for summer and most water is recycled. Our estimate: melting it takes only a little spare heat.

3

BREAKTHROUGH 3 · MOUNT AURORA

# A Snow Mountain You Can Walk Inside

*Our big idea: the mountain is the building, with warm pools and saunas inside a snow peak.*



*Cutaway view: four spa levels inside the snow peak, crowned by the glass Aurora Eye.*

## IN BRIEF

### THE PROBLEM

Long, dark winters are linked to seasonal affective disorder, a winter depression [12]. People need a warm, bright place to meet.

### THE IDEA

Four warm spa levels inside an insulated snow peak, heated by a hot spring.

### WHAT'S NEW

Alaska's Chena Hot Springs makes its own geothermal power and grows vegetables in greenhouses [13]. We turn a whole snow peak into an insulated building that is also a shelter.

### THE IMPACT

Rest and light for residents, income from visitors and a refuge in a disaster.

MOUNT AURORA · LEVEL 1

# Inside the Mountain

*Level 1 is the arrival floor: a welcome hall, changing rooms and the Mineral Lake grotto.*



WHERE THE WORLD IS TODAY

Iceland's Blue Lagoon bathes visitors in water from a geothermal power plant [14]. Mount Aurora goes further: the peak itself is the building.

REALITY CHECK

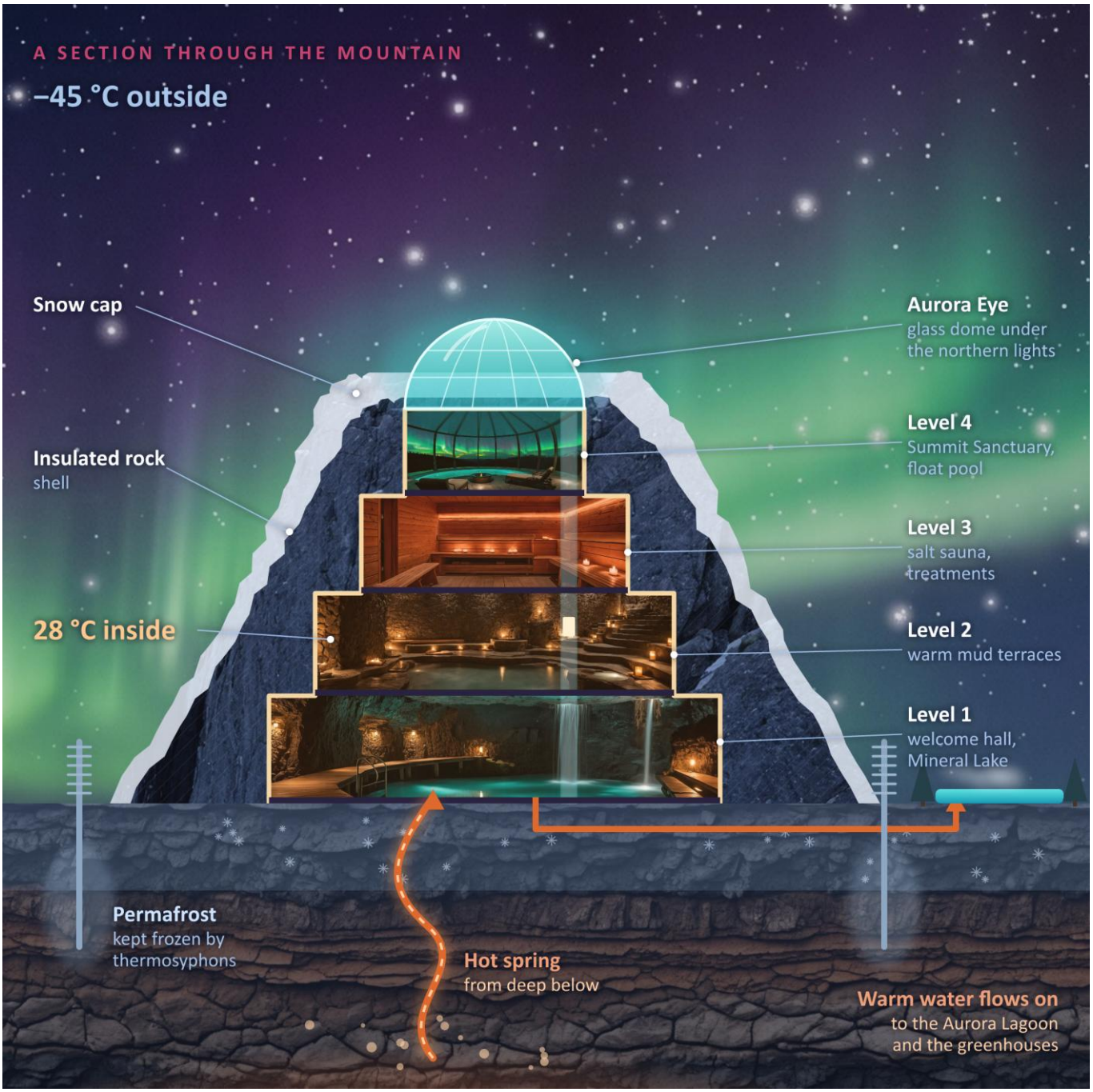
## 28 °C inside

An insulated rock shell keeps the spa warm at −45 °C. The hot spring must still be confirmed by drilling, and thermosyphons keep the ground frozen [15].



*The Mineral Lake grotto and its larch footbridge.*

# Four Floors of Warmth




## How it stays warm

A hot spring heats the pools, and the rock and snow hold the warmth in, like an igloo.

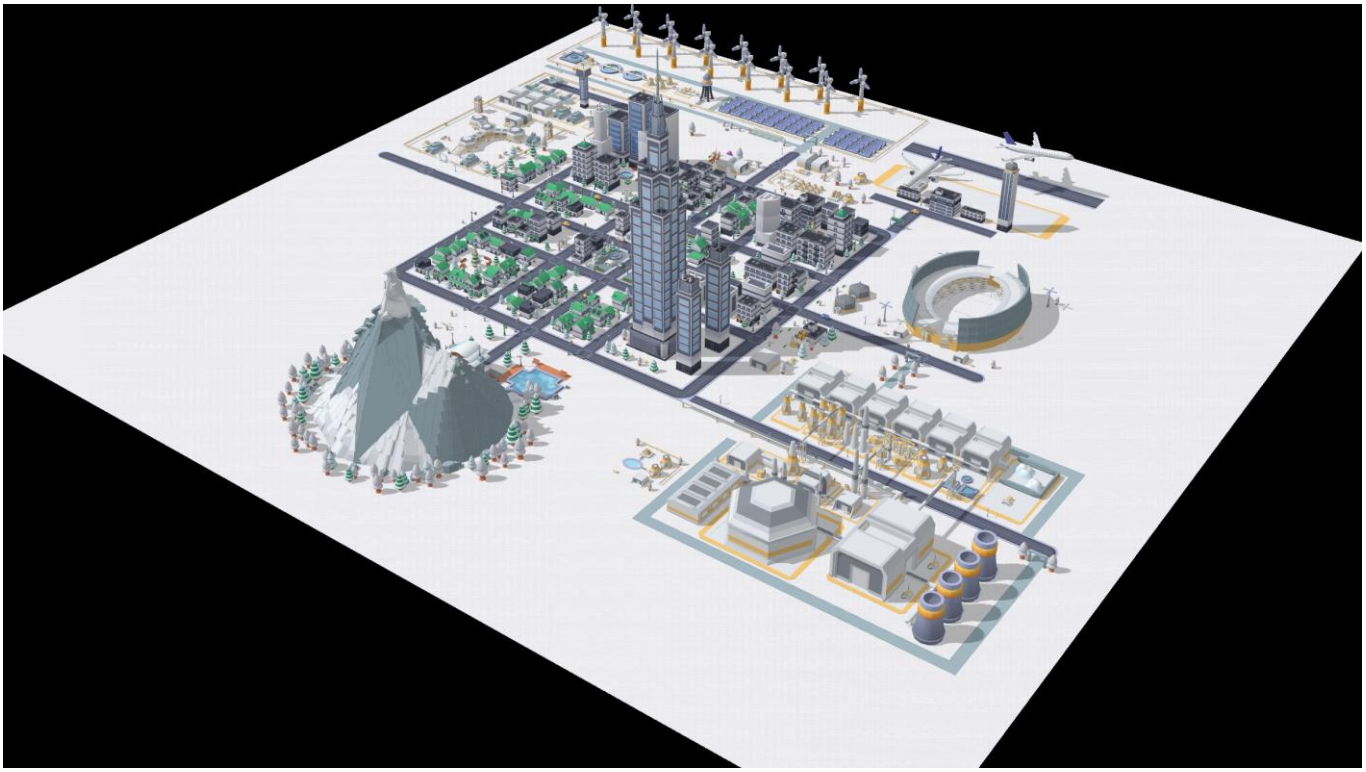
## Also a shelter

In a blizzard or a power cut, its heat comes from the ground, not the grid.

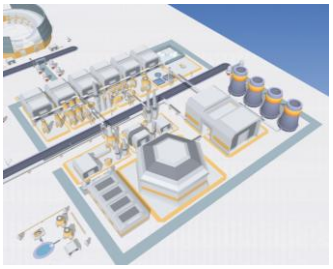
OUR ECO-POLIS MODEL

# Built by Us

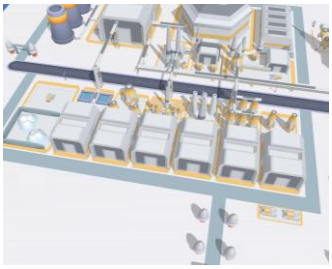
*Before any picture, we built Aurora Haven ourselves in the Eco-Polis Builder, place by place.*



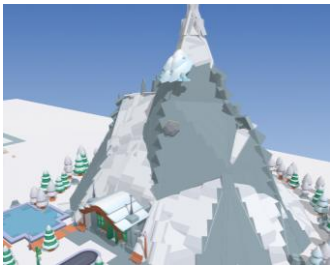
*The whole city in our Eco-Polis model.*



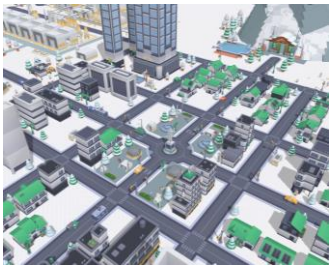
*Fusion Power Station*



*Snow-to-Food & Water Plant*



*Mount Aurora*



*Downtown*

**1 · Download our map**

Aurora_Haven.json

**2 · Open Eco-Polis**

ecopolis3d.netlify.app

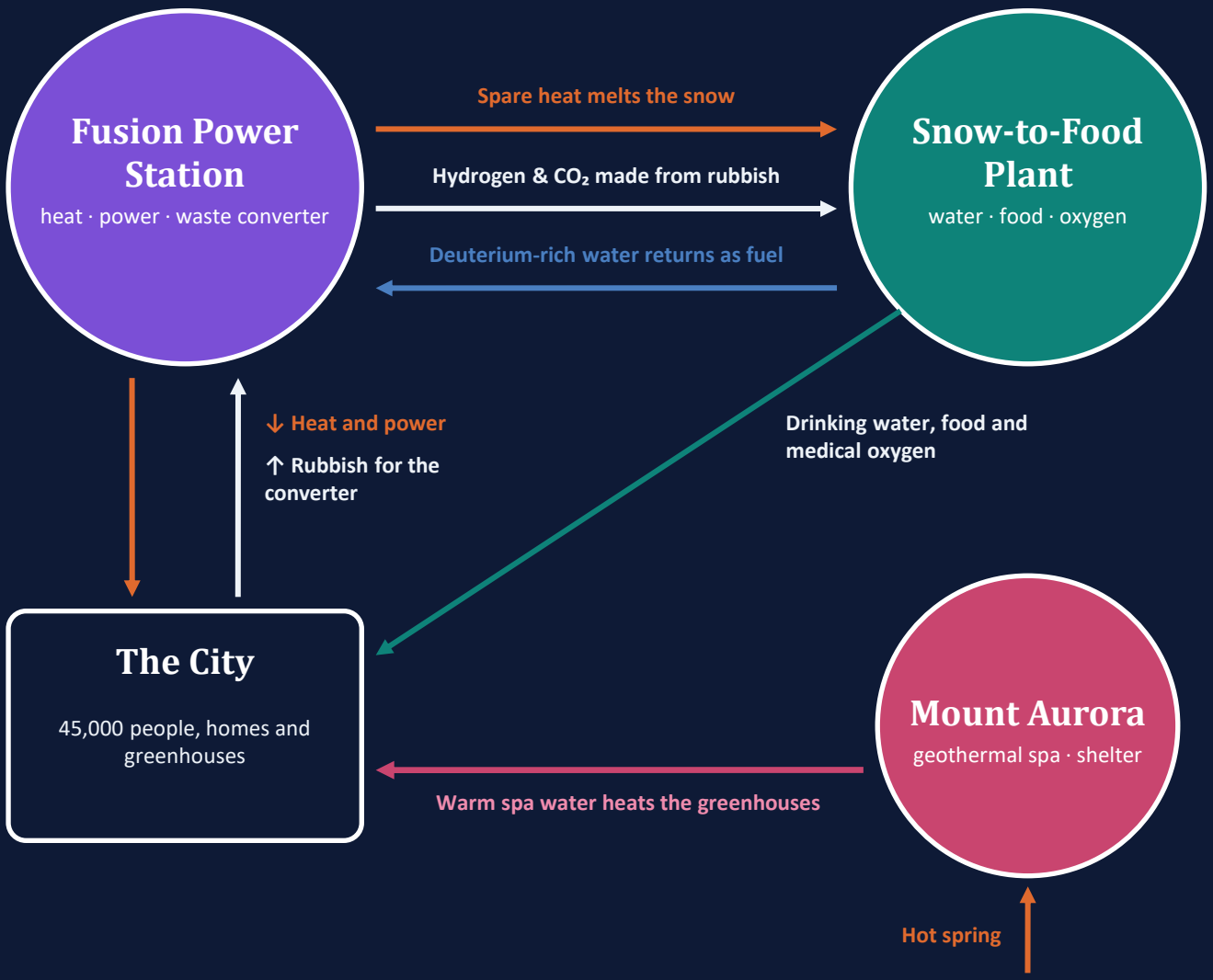
EXPLORE OUR 3D CITY

1. Scan code 1 to download our map file.
2. Scan code 2 to open the Eco-Polis Builder.
3. Click Load (the folder icon) and choose Aurora_Haven.json.
4. Click Lock (View), then Full Screen.
5. Drag to turn, scroll to zoom, click a building to read about it.

THE BIG IDEA

# Three Districts, One Loop

*Our three breakthroughs work as one system, so heat, water and waste go round.*




**Heat**

Spare fusion heat melts the snow: snow is the reactor's cooling tower.

**Water**

Deuterium-rich water returns as fuel; oxygen goes to the hospital.

**Waste**

Rubbish becomes hydrogen and $CO_2$ for food.

**Wellbeing**

Spa water heats the greenhouses; the mountain is a shelter.

EVERYDAY LIFE

# A Liveable City Every Day



*Snowpine Residences at dusk, sheltered by planted spruces.*

### HOMES

Family houses, eco-homes and high-rise apartments.

### LEARNING

Kindergartens, two schools and Arctic University.

### HEALTH

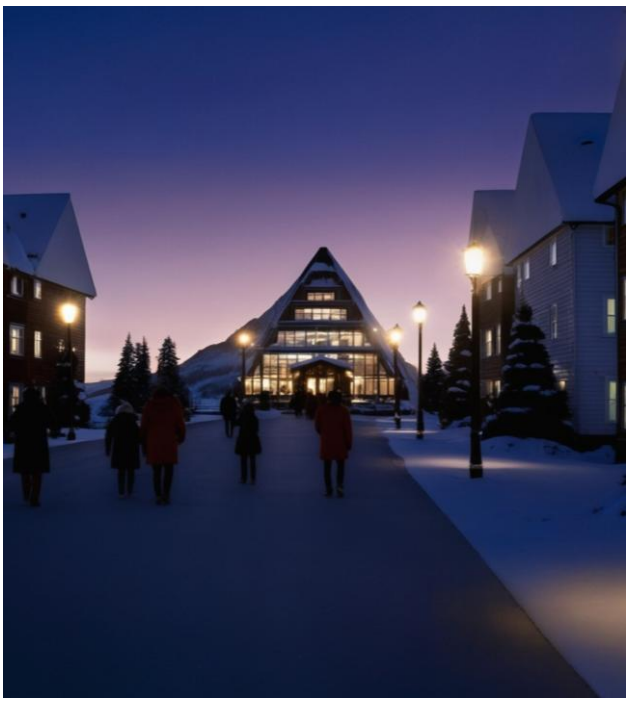
Snowflake General Hospital, open around the clock.

### SHOPPING

A supermarket, a night market and a mall with an indoor park.

### CULTURE AND GREEN SPACE

A cinema, an all-night library, the Snow Stadium and parks.



*The Mount Aurora Approach: low buildings and soft lamps keep the aurora in view.*



*The western eco-homes at dusk, with the wind farm behind.*

HOUSING

# Homes for the Polar Night

*Every home stays warm and bright at −45 °C without thawing the ground beneath it.*

## HOW A HOME BEATS THE COLD

### Insulated piles

Homes stand on piles above an air gap, so their warmth does not reach the permafrost.

### Super-insulated shell

Thick walls and triple glazing keep the heat in.

### Heat-recovery air

Outgoing air warms the fresh air coming in.

### Winter gardens

Glazed balconies add a warm buffer.

### Snow-shedding roofs

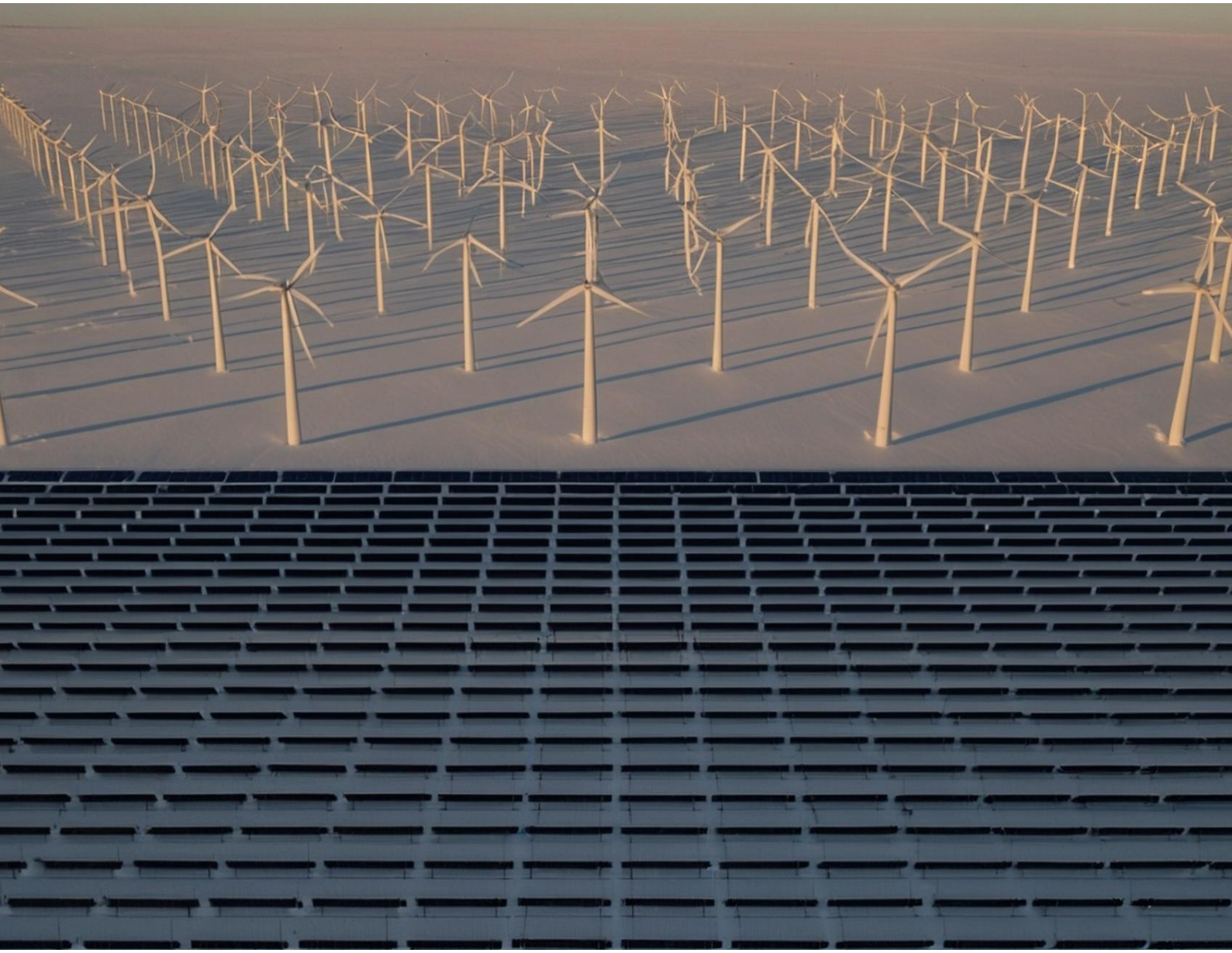
Steep roofs let heavy snow slide off.

### Light in the dark

Light tubes and dawn lamps for the polar night.

**A HOME FOR EVERYONE** Family houses, apartment towers, co-living, and homes for students and older people, all close to the heating loop and the metro.

# Energy for the Polar Night



*The wind and solar farms of the Northern Renewable Park.*

*Wind and sun start the city's own sun. Once fusion runs, they stand guard.*

**1 · START-UP**

Wind and solar farms charge a battery bank that starts the reactor.

**2 · RUNNING**

Fusion then powers and heats the whole city, day and night.

**3 · BACKUP**

If the reactor stops, wind, sun, geothermal and stored heat keep basic heating, the hospital and the shelters going.

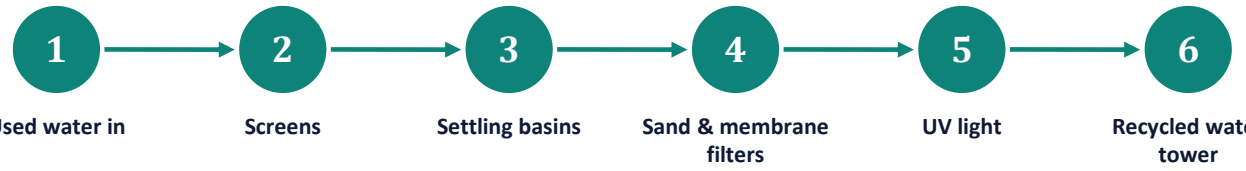
WATER AND WASTE

# Nothing Is Wasted



*The Water Recycling Line.*

## USED WATER, CLEANED IN SIX STEPS




Water from sinks and showers is cleaned in six steps and reused in toilets, laundries and greenhouses. Toilet water is treated on its own and becomes biogas.

The Circular Recycling Center sorts and repairs materials. Food scraps become biogas, and a plasma converter turns the rest into gas and building blocks.

**95%** of the city's water is used again

***Almost nothing goes to landfill in the permafrost.***



*The eco-tour reindeer herd on the tundra by the Snow Garden Park.*

URBAN ECO-SYSTEM

# Wild Neighbours

*We share the tundra: animals move through the city, and planted trees work as snow fences.*

## ANIMALS

**Wildlife corridors.** Tree belts link the tundra to the parks, with dim lights and underpasses.

**Reindeer.** Local herders keep a small eco-tour herd.

**Polar bears.** Fence sensors call a wildlife team when a bear comes close.

## PLANTS AND FARMED LIFE

**Living snow fences.** Belts of hardy spruce trap drifting snow and slow the wind.

**Aquaponics.** Fish feed lettuce and herbs in the greenhouse.

**Shelter and science.** Rock gardens shelter ptarmigans and hares.

**HOW THEY SURVIVE** Arctic foxes grow a thick winter coat that lets them survive down to about −50 °C [16]. The city gives them cover, safe crossings and dark nights.

TRANSPORT

# Getting Around Fossil-Free

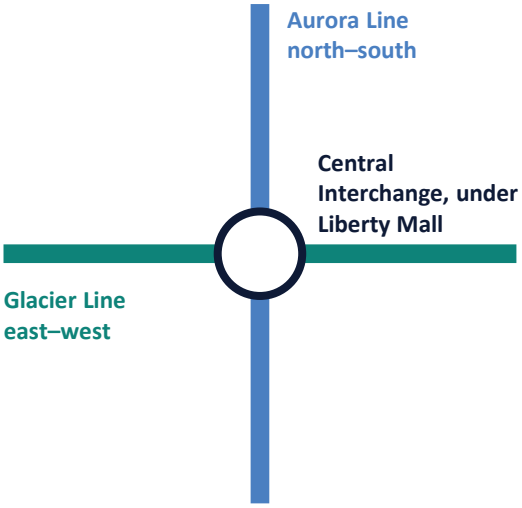


*Compact Polar Airport, the city's winter supply link.*

On the ground, everything is electric. Public transport starts with two driverless metro lines that meet under Liberty Mall, deep below the blizzards. Trams, buses and shared robotaxis stop at heated shelters.

Warm streets and tunnels sit on thick insulation, and thermosyphons, as on the Trans-Alaska Pipeline, keep the ground below frozen [15].

Planes fly on e-fuel made with clean power.




### Who rides

Commuters, students and fans, with room for wheelchairs and skis.

### How many

About 8,000 people an hour each way (our estimate).

### Safe in a storm

Insulated tunnels with doors straight into the shelters.

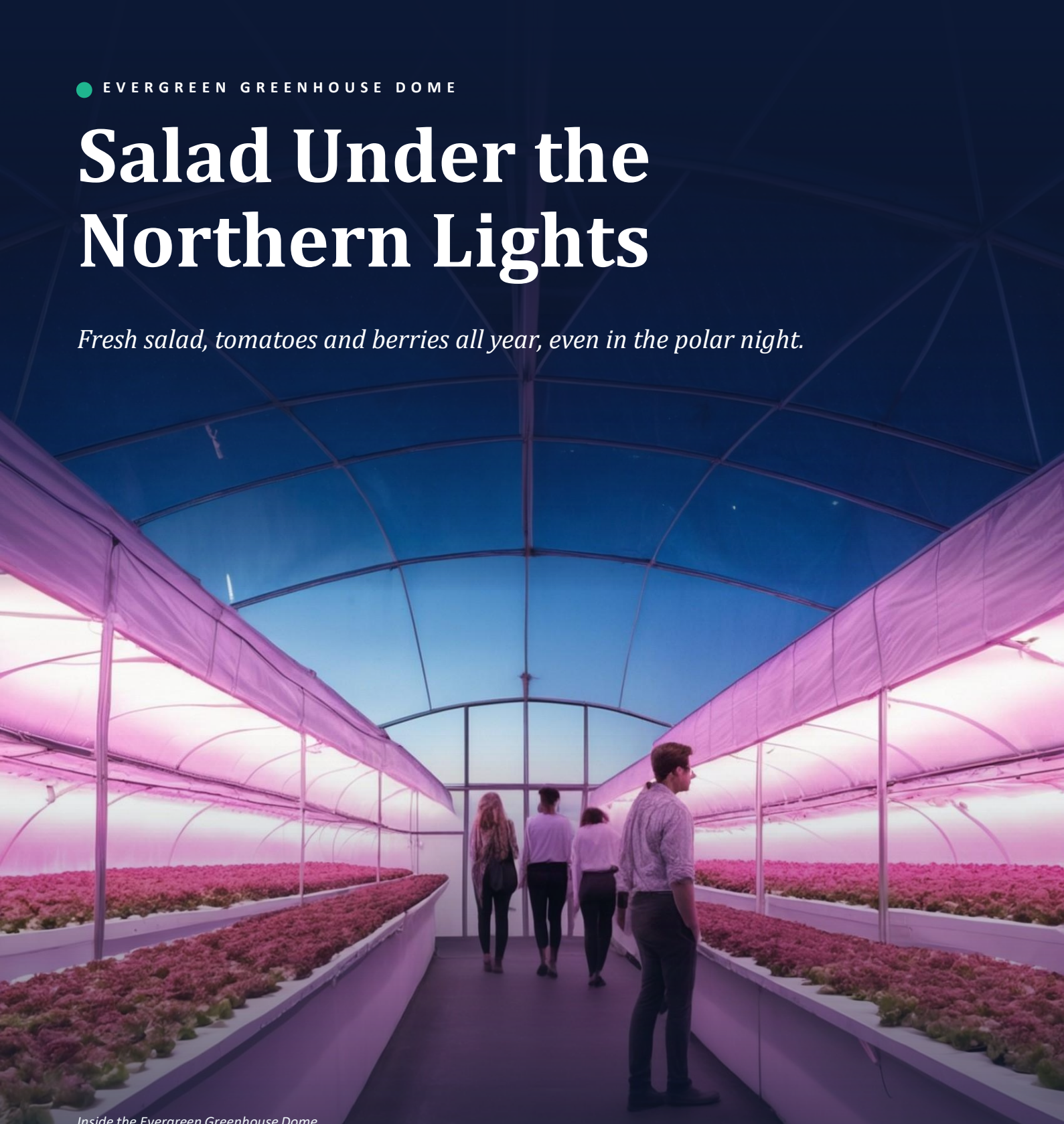
### Energy

All-electric; braking trains feed power back.

EVERGREEN GREENHOUSE DOME

# Salad Under the Northern Lights

*Fresh salad, tomatoes and berries all year, even in the polar night.*



*Inside the Evergreen Greenhouse Dome.*

The clear roof catches the midnight sun in summer. In the polar night, LEDs take over and thermal curtains keep the heat in, with warmth from the heating loop and Mount Aurora's spa water.

**≈ 90%**

less water than field farming [17]

SAFETY

# Safe and Ready for Any Storm



*The Polar Rescue and Resilience Station, north of downtown.*

The rescue station houses the fire brigade, ambulances, ice rescue and the 112 emergency centre, with a helicopter ready day and night.

Under Liberty Mall, four shelters hold supplies for every resident. Aurora Core, the city's AI control tower, sends blizzard warnings to every phone.

**30 days**

of supplies for every resident

**48 h**

warning before a blizzard

WEIGHING IT UP

# Pros & Cons

*No city is perfect: here is our honest balance sheet.*

## ADVANTAGES

**Zero carbon.** No fossil fuel burns anywhere.

**Closed loops.** Water and waste go round, not to landfill.

**Food security.** Tap water and much of the food are made here.

**Safety.** Shelters, blizzard warnings and a storm-proof metro.

**Wellbeing.** Everything a short walk away, plus a spa.

## LIMITATIONS AND RISKS

**Fusion is not ready.** ITER plans to burn fusion fuel only from 2039 [7].

**One heating loop.** If it fails, backup heat goes to the hospital and shelters first.

**Thawing ground.** A heat leak could crack foundations [3].

**Isolation.** In deep winter, supplies come only by air.

**High-tech risks.** AI systems need cyber-security; new foods need trust.

### COSTS

Fusion is one of the most expensive machines ever built: ITER alone is estimated at €18–22 billion [18]. Indoor farms save water but use far more electricity [17].

### TRADE-OFFS: WHAT WE CHOSE → WHAT WE GAVE UP

| | | |
|---|---|---|
| **Compact blocks that save heat** | → | smaller homes and less private space |
| **Metro, trams and shared robotaxis** | → | fewer private cars and less freedom to drive |
| **Lights that switch on only when needed** | → | darker streets, for the aurora and the wildlife |
| **Factory-made food** | → | no farmland needed, but more electricity and new tastes |

**OUR PLAN B** Build in stages: geothermal, wind, sun and storage first; fusion only once it is proven.



INVESTMENT OPPORTUNITY

# Invest in Aurora Haven

*The Mount Aurora Approach at dusk.*

*We are looking for partners to build a zero-carbon city, step by step.*

## THREE WAYS TO INVEST

**1 Green city bonds**

Fund Stage 1, repaid from heat and power sales.

**2 Community shares**

Residents co-own the wind and solar farms.

**3 Research partnerships**

Test new ideas with Arctic University.

## WHAT YOUR MONEY BUILDS, STAGE BY STAGE

1. **Stage 1 · A clean base:** geothermal, wind, sun, storage and the metro.
2. **Stage 2 · Food and water:** the Snow-to-Food plant, starting small.
3. **Stage 3 · Fusion:** only once fusion is proven (page 31).

## WHAT INVESTORS GET BACK

- Steady income from heat, power and food
- Visitors all year
- Know-how for other Arctic towns
- No fuel or carbon bills

*Future concepts, not tested projects: no return is guaranteed.*

**READY TO BUILD THE FUTURE?** Join Stage 1 with a green city bond.

EXPLORE

# Take the Tour: Eight Stops

*From downtown out to the plants, the snow peak and the windy north.*

1. **Founding Mayor Monument**
   The civic heart of the city.
2. **Central Interchange Station**
   Down to the metro and shelters.
3. **Snowflake General Hospital**
   The main hospital.
4. **Fusion Core**
   The city's own sun.
5. **Snow Harvest & Melt Hall**
   Where snow becomes water.
6. **Mount Aurora**
   The snow peak with a spa inside.
7. **Ice-Proof Wind Farm**
   Turbines built for storms.
8. **Polar Rescue and Resilience Station**
   The 112 emergency centre.



*Aurora Snow Stadium at dusk.*

# Places to Visit

*The numbers match the markers on the city plan on page 10.*

| | |
|---|---|
| 1 | Founding Mayor Monument |
| 2 | Aurora Civic Center & Municipal Hall |
| 3 | Snowflake General Hospital |
| 4 | Northern Lights School & Botanical Conservatory |
| 5 | Igloo Fresh Supermarket & Food Hall |
| 6 | Aurora Cinema & Performing Arts Center |
| 7 | Underground Metro & Emergency Shelters |
| 8 | Aurora Core – AI Control Tower |
| 9 | Aurora Haven Police Headquarters |
| 10 | Arctic University – Main Hall |
| 11 | Aurora Public Library |
| 12 | West Market Fire & Ambulance Station |
| 13 | Aurora Glacier Mall & Crystal Atrium |
| 14 | Snow Lantern Primary School |
| 15 | Pinecone Kindergarten |
| 16 | Aurora 81 – Polar Landmark Tower |
| 17 | Aurora Snow Garden Park |
| 18 | Mount Aurora – Snow Peak & Hanging Glacier |
| 19 | Summit Sanctuary & Aurora Eye |
| 20 | Aurora Lagoon – Outdoor Geothermal Mineral Pool |
| 21 | Fusion Core – Compact D-T Tokamak |
| 22 | Power Conversion & Heat Export Hall |
| 23 | Plasma Waste Converter |
| 24 | Snow Harvest & Melt Hall |
| 25 | Drinking-Water & Mineral Works |
| 26 | Bioreactor Silos |
| 27 | Ice-Proof Wind Farm |
| 28 | Bifacial Solar Farm |
| 29 | Water Recycling Line – Recycled Water Tower |
| 30 | Geothermal Power Plant |
| 31 | Polar Rescue and Resilience Station |
| 32 | Glass Terminal, Control Tower & Gate 1 Dreamliner |
| 33 | Evergreen Greenhouse Dome |
| 34 | Circular Recycling Center |
| 35 | Aurora Snow Stadium |



A DAY IN AURORA HAVEN · AN IMAGINED STORY

# Through a Resident's Eyes

*January, six weeks without a sunrise: a winter day for Mai, sixteen, a student here.*

**07:30** Mai's flat in the Aurora 81 Residences is warm without a heater. Breakfast is bread made from snow and air.

**08:10** The driverless metro takes her to school, blizzard or not.

**12:30** Lunch is a greenhouse salad. After school, she borrows a light-therapy lamp from the library.

**17:00** Every phone pings a blizzard warning two days ahead. Nobody panics.

**20:00** At Mount Aurora, she floats in the warm lagoon under the northern lights.

***"In the polar night, my city is the warmest place I know."***

*Walking home under the northern lights.*

RESEARCH INTEGRITY

# Sources & Assumptions

## HOW TO READ OUR NUMBERS

### Design values

Figures from our own plan, such as 45,000 residents, not from a real city.

### Our estimates

Our own sums, marked “our estimate”: reasonable, not exact.

### Future concepts

The fusion station and the food plant do not exist at this size yet. Nothing has been tested.

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**PICTURES** The cover and the city pictures are photo-realistic images made with AI tools (Stable Diffusion XL, and Nano Banana Pro for the cover sky) from renders of our own 3D model. The maps and diagrams are drawn from our model, and the Eco-Polis photos on page 21 are real screenshots.

**AI USE** We used AI tools to help us build the 3D model of the city and make the pictures, plans and diagrams in this magazine, and to check our spelling and fix logic errors in the text.

JOIN US

# Warm, Green and Ready

As we conclude this feature on Aurora Haven, we are left with more than just an impression of futuristic design – we are reminded of our shared responsibility toward the planet. The project clearly demonstrates that advanced technology and nature can coexist in harmony when guided by vision and innovation. Aurora Haven not only sets a precedent for resilient urban planning in extreme climates, but also serves as a powerful inspiration for a sustainable future where human communities are built to endure. May the concepts presented here spark further dialog and inspire bold solutions for generations to come.

**Be one of the first polar citizens.** Buy a community share, study at Arctic University, or spend a night at the Mount Aurora spa.

## AURORA HAVEN

*A polar city that thrives*

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SCAN TO READ

### The interactive edition

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